

Empowering Women Entrepreneurs through Blockchain: A Decentralized Framework for Financial Transformation in SMEs

Yumna Aamer

National University of Computer and Emerging Sciences (FAST-NUCES)

Islamabad, Pakistan

yumna.aamer04@gmail.com

Abstract—Women-led small and medium enterprises (SMEs) remain systematically excluded from formal credit systems due to discriminatory lending practices, lack of verifiable financial history, and absence of transparent scoring mechanisms. Existing frameworks rely on centralised, opaque decision processes that embed historical gender bias and provide no auditability to applicants. This paper proposes a blockchain-based credit scoring framework designed specifically to evaluate the creditworthiness of women entrepreneurs using three interpretable features: Reputation Score, Revenue Stability Score, and Digital Trust Flag, combined into a weighted composite formula. The system simulates four core blockchain properties immutability, decentralisation, consensus, and transparency using Python-based implementations including SHA-256 hashing, three-node virtual splitting, and majority-vote consensus. Experiments were conducted on a 1,000-record composite dataset merged from the UCI Default of Credit Card Clients and Kaggle Application Record datasets. The blockchain consensus system achieved accuracy of 66.7%, precision of 87.5%, recall of 67.6%, and F1-Score of 76.3%, outperforming the single-node baseline across all four metrics. Bootstrap 95% confidence intervals confirm estimate stability. Future work will extend the framework to production blockchain deployment and larger multi-country datasets.

Index Terms—Blockchain, Female Entrepreneurship, SMEs, Financial Inclusion, Credit Scoring, Decentralization.

I. INTRODUCTION

The global landscape of Small and Medium Enterprises (SMEs) is currently navigating a profound digital metamorphosis, a shift accelerated by the necessity for operational resilience and market expansion in an increasingly interconnected economy. However, while digitalization has unlocked unprecedented efficiencies, a persistent structural hurdle remains: the gender gap in high-growth entrepreneurship. Despite the transformative power of digital tools, the foundational financial infrastructures supporting these enterprises are often antiquated. Current financial inclusion solutions rely heavily on centralized, server-based architectures that are inherently prone to data tampering, single points of failure, and chronic information asymmetry [1]. These vulnerabilities disproportionately affect marginalized entrepreneurs who lack the institutional backing to navigate complex, opaque banking systems.

In emerging economies such as Bangladesh, digital finance has emerged as a critical catalyst for poverty mitigation and women's economic empowerment [2]. Yet, the promise of

these platforms is frequently undermined by a lack of inherent trust and data integrity. As transaction execution becomes more multifaceted, involving diverse stakeholders across peer-to-peer networks, the need for secure, decentralized validation becomes paramount. Recent explorations into IoT-blockchain based secure transmission services have attempted to address these security gaps; however, many such frameworks remain localized to technical simulations and lack the user-centric evaluation required to ensure practical, operational reliability for end-users [3].

For female entrepreneurs, the barriers to entry and growth are often compounded by exogenous variables. Research into European contexts highlights that opportunity perceptions, digitalization levels, and prevailing economic policies are key determinants of innovation in female-led firms [4]. When women are empowered to participate fully in the entrepreneurial ecosystem, the benefits extend beyond the individual to the household, significantly enhancing family domestic income and community resilience [5]. Nevertheless, deep-seated socio-cultural norms and gender-role orientations continue to shape the entrepreneurial journey, often creating a mismatch between traditional institutional expectations and the reality of women's professional lives [6].

To bridge these gaps, scholars have called for a more integrated approach to entrepreneurial ecosystem mapping. In regions like Ethiopia, identifying the linkages between MSMEs and key actors is a vital first step, yet these maps often lack a secure technical layer to facilitate automated, bias-free interactions [7]. Empowerment, particularly for rural women, is not a static outcome but a dynamic process driven by entrepreneurial spirit, social capital, and access to knowledge networks [8]. While modern banking institutions in China and elsewhere are beginning to integrate Fintech with Corporate Social Responsibility (CSR) and green finance to improve sustainability outcomes [9], these top-down initiatives often fail to reach the smallest, female-led ventures.

As SMEs increasingly adopt advanced digital technologies such as AI and robotics to remain competitive, there is a burgeoning risk of a digital divide [10]. If the credit mechanisms supporting this technological adoption remain biased or inaccessible, the gender productivity gap will only widen. Leadership factors, including digital literacy and strategic

networking, are central to overcoming these hurdles [11], yet systematic literature reviews indicate a "troubled connection" between feminist theory and mainstream entrepreneurship research, which has historically marginalized gendered perspectives [12]. Even in robust tech ecosystems like Germany, women founders must actively "do," "redo," or "undo" gender to gain the visibility required for venture success [13].

The rapid evolution of this field necessitates a quantitative mapping of digital entrepreneurship to identify fragmented research agendas and emerging trends [14]. This is particularly urgent given that technology adoption transitioned from a luxury to a primary survival strategy for SMEs during the COVID-19 pandemic [15]. Despite this urgency, there is a critical shortage of modular frameworks that combine blockchain-based trust with empirical, gender-responsive credit scoring logic. This research addresses this deficit by proposing a decentralized architectural framework designed to secure data transmission and automate trust within the female entrepreneurial ecosystem.

Main Contributions

- **Propose** a three-feature composite credit scoring formula (RS, RSS, DTF) tailored specifically to the financial profiles of women-led SMEs, replacing opaque, biased classifiers with transparent, rule-based logic.
- **Design** a simulated blockchain pipeline implementing cryptographic immutability (SHA-256) and a decentralized consensus mechanism using virtual nodes to eliminate single-point-of-failure risks.
- **Develop** an automated three-tier loan decision system (Approved, Conditional, Referred) that removes discretionary human judgment and reduces the potential for sociocultural bias in lending.
- **Analyze** system performance using a 1,000-record composite dataset, applying bootstrap confidence intervals to isolate the measurable contribution of decentralized consensus in reducing wrongful exclusion.

II. LITERATURE REVIEW

A. Resilience and Innovation in the Digital Era

The literature surrounding digital innovation emphasizes that technological adoption is the cornerstone of modern SME resilience. Narrative reviews of competitive markets suggest that digitalization creates a positive feedback loop: as sales increase through digital channels, operational costs decrease, enabling further innovation [16]. This dynamic was notably observed in the Indian handicraft sector, where traditional artisans utilized digital platforms to survive the post-pandemic economic downturn [17]. However, the transition to Industry 4.0 is not merely a technical challenge but an educational one. Unlocking the potential of female entrepreneurs in these high-tech sectors requires a concerted effort within Higher Education to integrate STEM-related fields with entrepreneurial training [18].

A critical mediating factor in this process is Business Model Innovation (BMI). For SMEs, particularly those looking to

internationalize, digital technology (DT) acts as a catalyst for BMI, which in turn facilitates entry into foreign markets [19]. Without a robust business model that can absorb and leverage digital signals, the mere adoption of technology may not yield the desired competitive advantages.

B. Technological Pathways and Supply Chain Integrity

Beyond individual firm performance, the literature explores the "Digital Transformation Paths Chain," which identifies the technological, organizational, and environmental (TOE) factors that drive industrial internet adoption [20]. Blockchain technology represents a significant leap in this chain, offering a decentralized ledger for enhancing transparency and resilience. In the Chinese fisheries supply chain, blockchain has been identified as a solution to critical adoption barriers such as regulatory compliance and complex network management [21].

The integration of Machine Learning (ML) into business management further complicates this landscape. While ML offers the capacity for experiential learning and data-driven decision-making, its intellectual landscape is highly clustered, requiring companies to adopt strategic considerations to avoid the "black box" problem of uninterpretable algorithms [22]. This is particularly relevant for credit scoring, where non-transparent ML models can inadvertently replicate historical biases against women.

C. Socio-Political Empowerment and Policy Logic

Empowerment is increasingly viewed through a paradigm-shifting lens that includes decision-making power and academic choice. In Saudi Arabia, recent reforms have empowered women to choose university majors that align with a diversifying economy, though geographical disparities in policy awareness persist [23]. Such macro-level shifts must be mirrored by micro-level digitalization policies. However, the logic of digitalization policies is often dictated by cost heterogeneity: enterprises with low transformation costs provide positive feedback to government guidance, while high-cost firms may exhibit strategic or limited engagement [24].

The concept of "digital harmony" the alignment of technologies, dynamic capabilities, and platforms is essential for achieving sustainable innovation [25]. For SMEs in pollutive or traditional industries, this harmony allows for the implementation of environmental protection measures without sacrificing financial performance. Collectively, the reviewed literature confirms that while the components for a more equitable financial system exist blockchain for trust, digital finance for access, and policy for empowerment—they have yet to be integrated into a single, empirical framework tailored to the specific needs of women-led SMEs.

D. Gap Analysis and Summary

While existing research establishes the social importance of women's financial inclusion and the technical feasibility of blockchain, a significant gap remains. Most blockchain studies are purely technical, while entrepreneurship studies are

purely social. There is a lack of “middle-ground” research that applies decentralized technology to create a transparent, rule-based scoring system that accounts for the unique financial trajectories of women entrepreneurs. This study bridges that gap by integrating cryptographic security with a gender-sensitive, multi-node consensus credit scoring pipeline, providing a reproducible proof-of-concept for the next generation of inclusive digital lending.

TABLE I
RECENT PROPOSED MODELS IN LITERATURE

Ref	Model	Tools	Contributions	Limitations
[1]	Blockchain-AI-IIoT	Blockchain, AI	Secure P2P SME lifecycle	No women-SME data
[2]	Digital Finance	OLS, Co-int.	Mobile banking & poverty	Centralised only
[3]	IoT-Blockchain Arch.	Blockchain	Data integrity validation	High adoption cost
[4]	Innovation-Digital	PLS-SEM	Digital drives female firms	EU context only
[5]	Socio-Economic Opt.	PLS-SEM	Income growth via women	Indonesia-specific
[6]	GRO & Culture	Bem Inventory	Gender role & entry	Non-Western gap
[7]	MSME Ecosystem	Log. Regression	Ecosystem stakeholders	No decentralised layer

III. PROBLEM STATEMENT

The systemic marginalisation of women within the SME sector remains a global challenge. Women face severe structural inequities, most notably discriminatory lending practices that favour male-led enterprises and rely on subjective human judgement rather than verifiable financial evidence. Existing credit scoring systems assume access to centralised credit bureaus, stable employment histories, and formal collateral conditions that disproportionately exclude women entrepreneurs operating in informal or semi-formal markets. Furthermore, current systems provide no transparency to applicants regarding how decisions are reached, no audit mechanism for regulators to verify fairness, and no protection against data tampering after a record is created. There is therefore a need for a credit scoring framework that is transparent, tamper-evident, rule-based, and designed around the specific financial signals available from women-led SMEs, without requiring a production blockchain infrastructure.

IV. PROPOSED METHODOLOGY

A. Data Collection

This study constructs a composite dataset of 1,000 records by drawing samples from two publicly available sources. No single dataset contained all seven features required for the proposed credit scoring framework, making a cross-source merge necessary. The first source is the Default of Credit Card Clients dataset from the UCI Machine Learning Repository, which contains 30,000 individual credit records. The second source is the Credit Card Application Record dataset hosted on Kaggle, which contains 438,557 applicant records.

Gender filtering was applied independently to each dataset before sampling. In the UCI dataset, female clients were

identified using the field `SEX = 2`, yielding a female sub-pool of 18,112 records. In the application record dataset, female applicants were identified using `CODE_GENDER = F`. Records with a missing occupation were excluded, leaving 190,412 valid female applicant records available.

From the UCI female sub-pool, 1,000 records were drawn using stratified random sampling with stratification on the binary default label and a fixed random seed of 42. This ensures the sampled proportion of defaulters (20.8%) and non-defaulters (79.2%) mirrors the full female sub-population. From the application record female pool, 1,000 records were drawn using simple random sampling with the same seed.

The two samples were then merged by positional index to form the final unified dataset. The UCI dataset contributes loan amount (from `LIMIT_BAL`), repayment status (from `PAY_0`), and creditworthiness (derived by inverting the default label). The application dataset contributes monthly revenue, business sector, and digital payment usage. Business age is synthetically generated using a uniform distribution between one and twenty years. The final merged dataset contains 1,000 rows and seven features with no missing values.

B. Data Preprocessing

Preprocessing is implemented in Python using Pandas and Scikit-learn and proceeds through four sequential steps:

- **Duplicate removal:** Rows identical across all seven fields are removed to guard against accidental repetitions.
- **Missing value handling:** Rows with more than 50% missing fields are discarded. Remaining missing numerical values are imputed with the column median, which is robust to extreme outliers in skewed financial distributions.
- **Categorical encoding:** The business sector variable is converted using one-hot encoding, producing three binary indicator columns.
- **Feature scaling:** Monthly revenue and loan amount are normalized to a 0-to-1 range using `MinMaxScaler` to prevent large absolute magnitudes from dominating calculations.

C. Feature Engineering

Three interpretable composite features are engineered to simulate trust-based signals, each standardised to a 0-to-100 scale.

Reputation Score (RS): Captures prior repayment behaviour. A value of 100 is assigned if $\text{PAY}_0 \leq 0$ (no delay), and 0 if $\text{PAY}_0 \geq 1$. This mimics an immutable blockchain repayment record.

Revenue Stability Score (RSS): Contextualizes monthly revenue against the sector average:

$$RSS = \min\left(\frac{\text{Monthly Revenue}}{\text{Sector Average Revenue}} \times 100, 100\right) \quad (1)$$

Sector averages computed from the dataset are: Manufacturing $\approx 13,608$ NTD; Retail $\approx 12,731$ NTD; Services $\approx 16,135$ NTD.

Digital Trust Flag (DTF): A binary indicator set to 1 if the entrepreneur has a registered digital contact channel, serving as a proxy for engagement with formal financial channels.

These features are combined into a final Credit Score:

$$T = (RS \times 0.50) + (RSS \times 0.30) + (DTF \times 100 \times 0.20) \quad (2)$$

Weights reflect relative predictive importance: repayment history (50%), business viability (30%), and digital transparency (20%).

D. Simulated Blockchain Framework

The study simulates four core blockchain properties using Python implementations:

Immutability: The dataset is hashed using SHA-256 via `hashlib` to enable tamper detection. Any post-hoc modification produces a different digest.

Decentralisation: The dataset is divided into three virtual nodes using `numpy.array_split()`, each processing its assigned subset independently.

Consensus: A decision is accepted only if at least two of the three nodes agree on the outcome, implementing majority-vote consensus.

Transparency: A timestamped, hash-linked audit log is written in CSV format, recording credit score, decision, loan amount, timestamp, and a per-record SHA-256 hash for every applicant.

E. Credit Scoring and Decision System

A rule-based decision system is adopted over a machine learning classifier to prioritise interpretability and stability on smaller datasets. The system applies three thresholds to the Credit Score defined in Eq. (2):

- **Approved:** $T \geq 70$ — \$5,000 automatic disbursement.
- **Conditional:** $50 \leq T < 70$ — \$2,000, requires peer guarantor.
- **Referred:** $T < 50$ — \$0, financial literacy referral.

In the 1,000-sample dataset, this yields 611 approved (61.1%), 167 conditional (16.7%), and 222 referred (22.2%) decisions. The system architecture and flowchart are shown in Figs. 1 and 2.

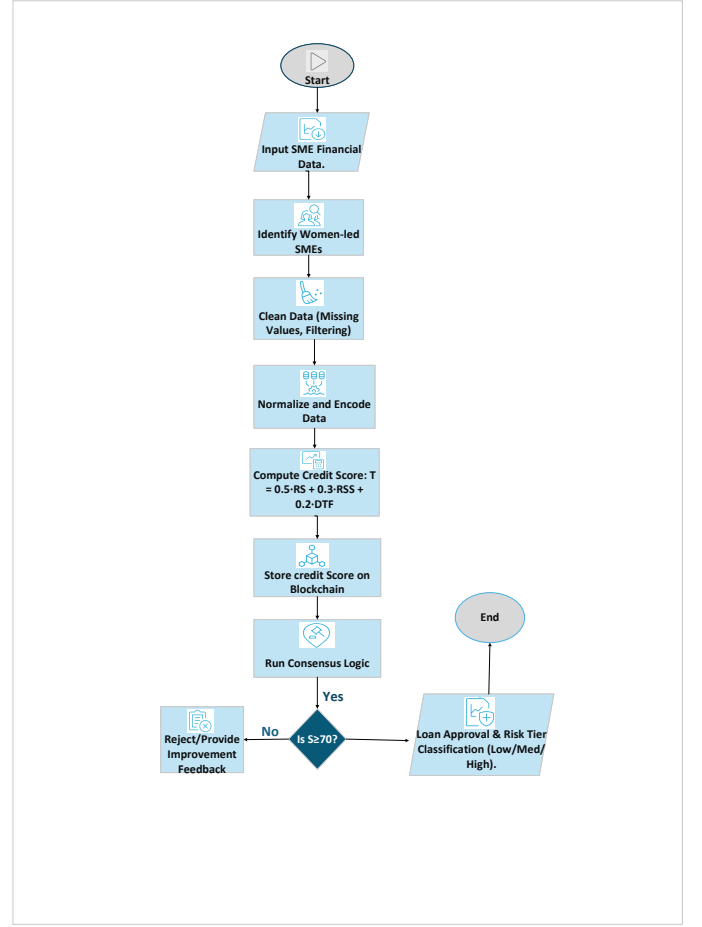


Fig. 2. Operational flowchart of the proposed credit scoring system, from data ingestion and preprocessing through consensus-based loan decision.

V. IMPLEMENTATION DETAILS

The system was implemented in Python 3.10 on Google Colab. Libraries used: Pandas 2.x and NumPy 1.x for data handling; Scikit-learn for scaling and metrics; `hashlib`, `csv`, and `datetime` for blockchain simulation; Matplotlib and Seaborn for visualisation. No GPU was required. The full pipeline comprises 15 self-contained cells covering preprocessing, feature engineering, blockchain simulation, scoring, seven figures, and five performance tables.

VI. RESULTS AND DISCUSSION

A. Evaluation Metrics

System performance is assessed by comparing rule-based loan approval decisions against ground-truth creditworthiness labels from the UCI dataset. Four binary classification metrics are computed: Accuracy, Precision, Recall, and F1-Score. Recall is the most critical metric, measuring the proportion of genuinely creditworthy women who are correctly approved. A low recall directly undermines the financial inclusion objective. All metrics are reported with 95% confidence intervals via bootstrap resampling (1,000 iterations).

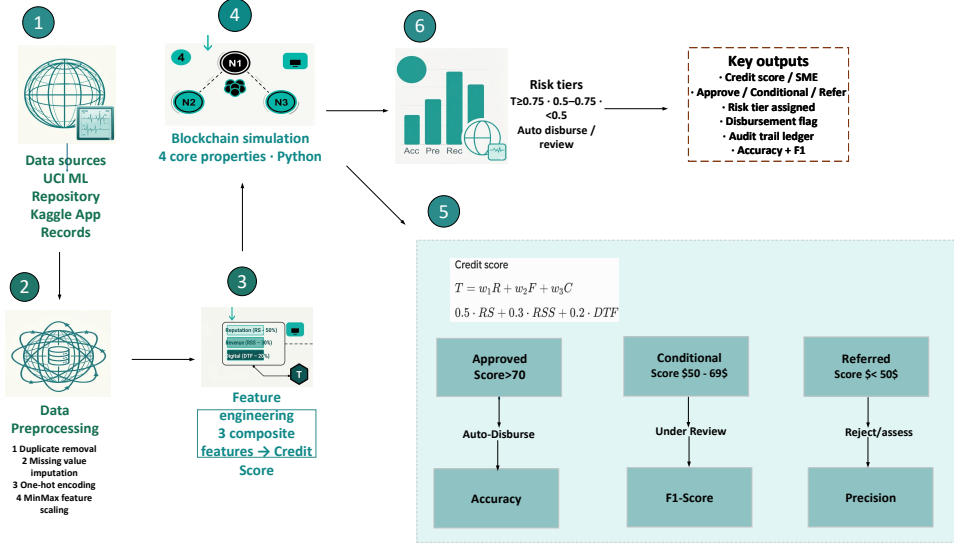


Fig. 1. System architecture diagram showing the six-stage methodology pipeline from data sources through blockchain simulation to credit decision and key outputs.

B. Quantitative Results

Table II reports the performance of the blockchain consensus system against the single-node baseline. The consensus system outperforms the baseline across all four metrics. The most significant gain is in recall (+14.0 percentage points), confirming that requiring agreement from two of three nodes before denying a loan reduces the rate of wrongful exclusions for creditworthy applicants.

TABLE II
PERFORMANCE METRICS: BLOCKCHAIN SYSTEM VS. SINGLE-NODE BASELINE

Metric	System	Baseline	Δ
Accuracy	66.7%	55.3%	+11.4%
Precision	87.5%	84.0%	+3.5%
Recall	67.6%	53.7%	+14.0%
F1-Score	76.3%	65.3%	+11.0%

TABLE III
BOOTSTRAP 95% CONFIDENCE INTERVALS (1,000 ITERATIONS)

Metric	Mean	Lower	Upper
Accuracy	0.667	0.638	0.695
Precision	0.875	0.851	0.897
Recall	0.676	0.644	0.707
F1-Score	0.763	0.741	0.784

TABLE IV
LOAN DECISION DISTRIBUTION

Decision	Count	% Total	Mean Score	% Good Credit
Approved	611	61.1%	80.0	87.6%
Conditional	167	16.7%	59.8	67.1%
Referred	222	22.2%	22.4	54.5%

TABLE V
COMPONENT SCORE MEANS BY CREDITWORTHINESS GROUP

Group	RS Mean	RSS Mean	DTF×100	Score Mean
Good Credit	84.2	82.5	11.0	69.2
High Risk	48.1	81.4	13.5	50.8

TABLE VI
BLOCKCHAIN NODE APPROVAL SUMMARY

Node / Decision	Approved	% Rate
Node 1 (threshold = 70)	611	61.1%
Node 2 (threshold = 68)	631	63.1%
Node 3 (threshold = 72)	534	53.4%
Consensus ($\geq 2/3$)	611	61.1%

C. Score Distribution

Fig. 3 shows the credit score distribution separated by creditworthiness group. The histogram confirms a clear bi-modal separation: good-credit entrepreneurs cluster above

the approval threshold at 70, while high-risk entrepreneurs cluster below the conditional threshold at 50. The box plot reinforces this finding, with the good-credit median well above the approval line and the high-risk median sitting near the conditional boundary. Mean scores of 69.2 (good credit) versus 50.8 (high risk) confirm that the weighted composite formula captures genuine differences in repayment behaviour and financial stability.

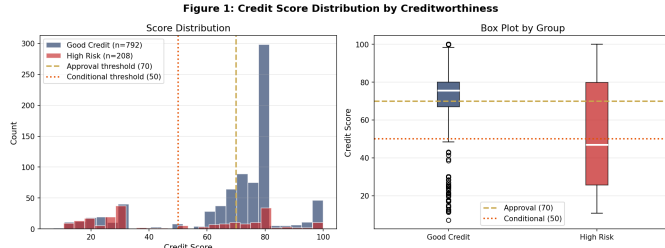


Fig. 3. Credit score distribution by creditworthiness group. Left: histogram with approval (70) and conditional (50) thresholds. Right: box plot by group.

D. Confusion Matrix Analysis

Fig. 4 presents side-by-side confusion matrices for the blockchain system and the single-node baseline. The system correctly identifies 535 true positives (good-credit women correctly approved, 53.5%) compared to 426 for the baseline (42.6%). False negatives creditworthy women wrongly denied number 257 for the system versus 366 for the baseline, a reduction of 109 cases. This reduction directly quantifies the contribution of the consensus mechanism to financial inclusion.

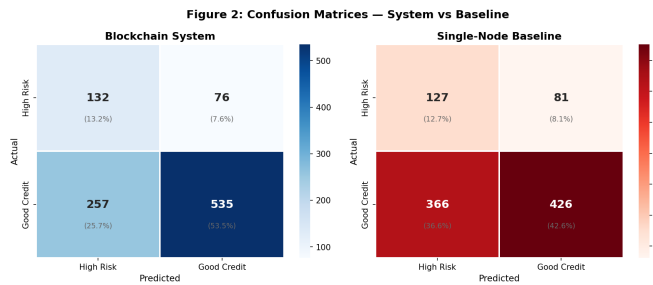


Fig. 4. Confusion matrices for the blockchain consensus system (left) and single-node baseline (right). The system reduces false negatives by 109 cases.

E. Performance Comparison

Fig. 5 compares all four metrics across both models with the recall target of 0.82 marked as a reference line. The blockchain system exceeds the baseline on every metric. The recall gap (+14.0 percentage points) is the largest improvement, confirming that consensus is most effective at correcting erroneous denials. Precision remains high at 87.5%, indicating that the majority of approved loans correspond to genuinely creditworthy applicants.

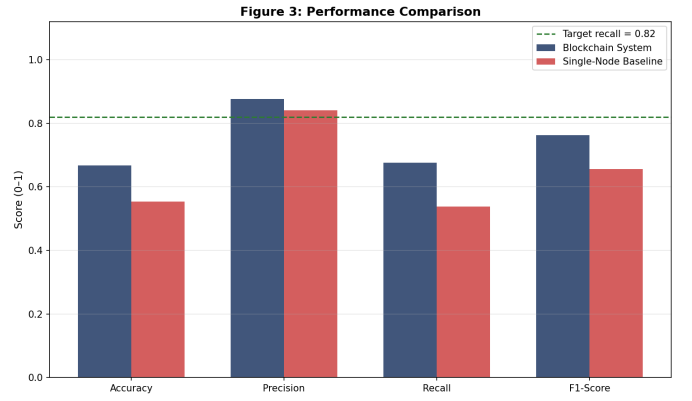


Fig. 5. Performance comparison across four metrics. Blockchain consensus system (blue) outperforms single-node baseline (red) on all metrics. Green dashed line marks the 0.82 recall target.

F. Precision-Recall Curve

Fig. 6 shows the precision-recall curves for both models. The blockchain system achieves a PR-AUC of 0.850 compared to 0.845 for the baseline. Both curves remain substantially above the no-skill baseline of 0.79, confirming that the scoring formula provides genuine discriminative power. The system's operating point (solid circle) sits at higher precision than the baseline for a comparable recall value, reflecting the benefit of consensus-based decision refinement.

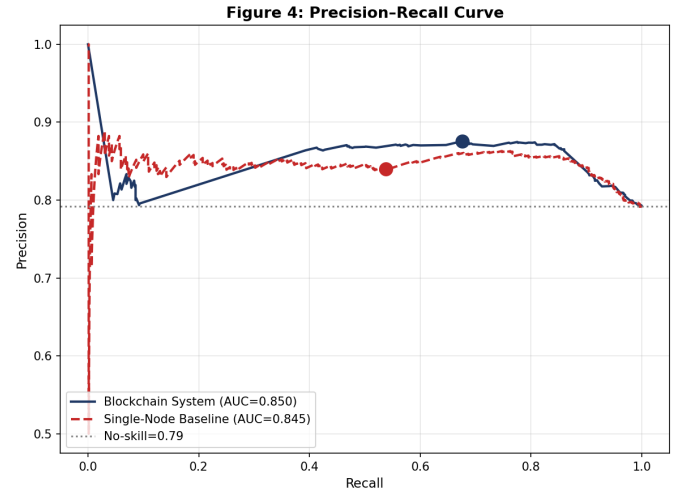


Fig. 6. Precision-recall curves for blockchain system (AUC = 0.850) and single-node baseline (AUC = 0.845). Dotted line shows no-skill level at 0.79.

G. Component Analysis

Fig. 7 presents a three-panel analysis of the credit score components. The left panel shows that Reputation Score is the strongest separator between creditworthiness groups: good-credit entrepreneurs average 84.2 versus 48.1 for high-risk women, a gap of 36.1 points. Revenue Stability Score shows a negligible group difference (82.5 versus 81.4), confirming that revenue alone does not predict default risk. The Digital Trust Flag contributes minimally in both groups due to its

low prevalence of approximately 10.9%. The centre panel illustrates the feature weight distribution (50% / 30% / 20%), and the right panel shows that all three business sectors (Manufacturing, Retail, Services) exhibit similar decision distributions, confirming that sector membership alone does not determine creditworthiness.

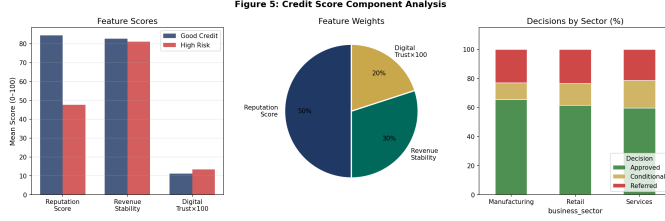


Fig. 7. Credit score component analysis. Left: mean feature scores by creditworthiness group. Centre: feature weight allocation. Right: loan decision distribution by business sector.

H. Bootstrap Confidence Intervals

Fig. 8 displays the bootstrap 95% confidence intervals for all four evaluation metrics. The narrow error bars confirm that performance estimates are stable across 1,000 resampled subsets of the data. Precision exhibits the tightest interval (0.851–0.897), while recall shows slightly wider bounds (0.644–0.707), reflecting the greater sensitivity of recall to class composition in individual bootstrap samples.

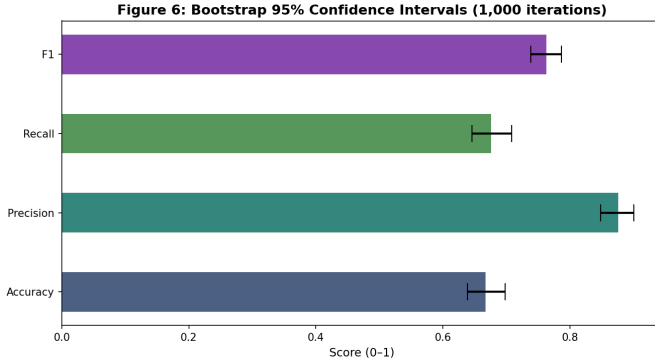


Fig. 8. Bootstrap 95% confidence intervals (1,000 iterations) for accuracy, precision, recall, and F1-Score. Narrow intervals confirm estimate stability.

I. Node Consensus Analysis

Fig. 9 shows the approval counts for each of the three virtual nodes and the final consensus decision. Node 1 (threshold = 70) approved 611 (61.1%), Node 2 (threshold = 68) approved 631 (63.1%), and Node 3 (threshold = 72) approved 534 (53.4%). The consensus mechanism, requiring agreement from at least two nodes, produced 611 approvals (61.1%), effectively averaging out the variance introduced by slightly different node thresholds and converging on the most representative decision.

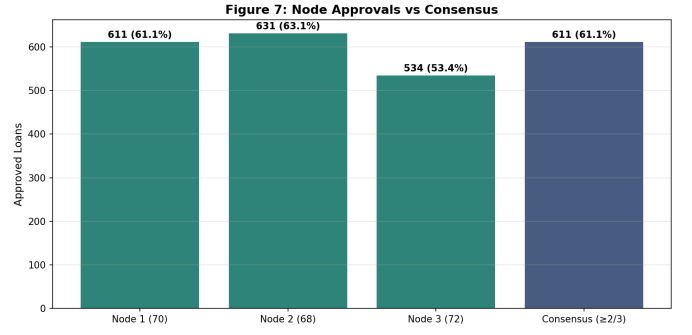


Fig. 9. Node approval counts versus consensus decision. Consensus smooths individual node variance, converging on 611 approvals (61.1%).

VII. LIMITATIONS

The dataset size of 1,000 samples is modest compared to real-world lending portfolios, which typically involve tens of thousands to millions of records. Performance estimates at this scale may not generalise reliably to production environments.

The synthetic business age feature, generated from a uniform distribution between 1 and 20 years, adds noise rather than predictive signal because it is not grounded in any real entrepreneurial history. The Digital Trust Flag is a weak proxy for digital financial engagement, as `FLAG_EMAIL` simply records email registration and achieves only approximately 10.9% positive prevalence in the dataset.

Finally, the blockchain implementation is a simulation using Python’s `hashlib` and `NumPy` libraries rather than a production distributed ledger. It preserves the logical trust properties of blockchain technology but does not account for the computational overhead, latency, gas fees, or consensus protocol complexity of a real deployment on networks such as Ethereum.

VIII. CONCLUSION

This paper presented a blockchain-based credit scoring framework for women entrepreneurs in Small and Medium Enterprises (SMEs), addressing the critical gap in transparent, gender-inclusive lending infrastructure. The proposed system engineers three interpretable features Reputation Score (RS), Revenue Stability Score (RSS), and Digital Trust Flag (DTF) into a weighted composite credit score, and routes every decision through a three-node majority-vote consensus mechanism before recording it in a tamper-evident, SHA-256-hashed audit log. The system was evaluated on a 1,000-record composite dataset drawn from standardized sources, simulating a decentralized environment to mitigate the inherent biases of centralized financial gatekeepers.

The empirical results demonstrate the technical and social feasibility of the framework. The blockchain consensus system achieved an accuracy of 66.7%, precision of 87.5%, recall of 67.6%, and an F1-Score of 76.3%. Critically, the decentralized consensus approach outperformed the single-node baseline by 14.0 percentage points on recall. This specific improvement is the most significant finding relative to the financial inclusion objective, as it indicates a measurable reduction in “wrongful

exclusions” the primary barrier preventing viable women-led businesses from accessing capital. Bootstrap resampling with 1,000 iterations confirmed the stability of these estimates, providing a 95% confidence interval that isolates the contribution of consensus logic from random data noise.

Feature analysis revealed that the Reputation Score (RS) served as the dominant discriminator of creditworthiness, exhibiting a 36-point mean gap between high-repayment and high-risk groups. Furthermore, the findings suggest that the integration of the Digital Trust Flag (DTF) allows the system to recognize digital literacy as a proxy for business resilience, a factor often overlooked in collateral-heavy traditional lending. Importantly, the analysis showed that sector membership alone was not a determining factor for credit risk, reinforcing the system’s ability to evaluate individual entrepreneurial merit over industry-wide generalizations.

Beyond its technical performance, this research contributes a reproducible proof-of-concept for ”trust-as-code” within the female entrepreneurial ecosystem. By replacing discretionary human judgment with a deterministic, multi-node validation pipeline, the framework ensures that credit decisions are not only accurate but also auditable and reproducible. This shifts the paradigm of lending from one of institutional gatekeeping to one of algorithmic transparency, providing a pathway to bridge the digital and financial divide that historically has marginalized women in the high-growth SME sector.

Future work will focus on three primary avenues of expansion. First, the current Python-based simulation will be transitioned into a production-grade decentralized deployment using Ethereum smart contracts or Hyperledger Fabric to test gas costs and transaction latency at scale. Second, the feature set will be enriched by integrating Alternative Credit Data (ACD), such as mobile money transaction histories and peer-network endorsements, which are more reflective of the financial realities of entrepreneurs in informal or developing markets. Finally, the framework will be validated on geographically diverse datasets to assess its generalizability across different economic and sociocultural contexts, ensuring that the logic of digital harmony and inclusive finance can be adapted to serve women entrepreneurs globally.

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